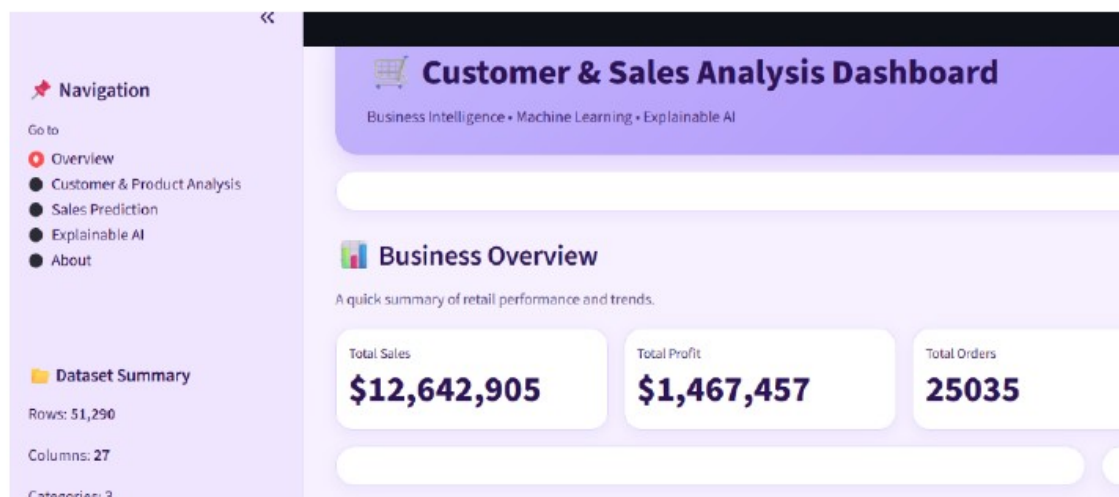


Business Intelligence Mini Project Report

Customer and Sales Analysis System for Retail Stores using Business Intelligence

AI-Powered Decision Support Platform for Retail Analytics

Built with: Python · XGBoost · SHAP · Streamlit



Submitted as per the mini-project report structure and adapted to match the reference BI report style.

1. Title of the Project

Customer and Sales Analysis System for Retail Stores using Business Intelligence

AI-Powered Decision Support Platform for Retail Performance Analysis, Sales Forecasting, and Explainable Decision-Making

Field	Details
Domain	Retail Analytics / Business Intelligence
Course	ITL601 — Business Intelligence
Technology	Python, Pandas, XGBoost, SHAP, Plotly, Matplotlib, Streamlit
Dataset	Retail Superstore Dataset
Type	Regression-based Predictive BI System
User Interface	Lavender-themed interactive Streamlit dashboard with KPI cards and analytical views

The proposed system is designed as an end-to-end BI solution. It does not stop at creating charts from historical data; instead, it combines descriptive analytics, predictive analytics, and explainable AI into one interface. The project is intended for retail managers, sales analysts, and business users who need both high-level performance summaries and detailed customer or product-level insights before taking action.

2. Abstract

Problem: Retail organizations often have rich transaction data, but without a BI system it becomes difficult to identify strong regions, profitable products, valuable customer segments, and future sales opportunities.

This project presents a Business Intelligence system for analyzing retail sales data and predicting future sales values using machine learning. The system is built on the Superstore retail dataset and integrates data preprocessing, feature engineering, interactive visualization, predictive modeling, and explainable AI in a single dashboard. The main predictive model used is XGBoost Regression, which is suitable because the target variable, sales, is continuous in nature. The dashboard is developed using Streamlit and displays a business overview with KPI cards, monthly sales trend, regional sales distribution, customer and product insights, sales prediction controls, and SHAP-based explainability panels.

The interface allows users to move from broad business questions to detailed analytical views. For example, a manager can first examine total sales and total profit, then drill into segment or region-level patterns, and finally test a set of category, sub-category, region, segment, quantity, discount, and profit inputs to estimate predicted sales. The project also improves decision transparency by showing feature importance and SHAP beeswarm plots so that business users can understand which variables most strongly influence predictions.

Overall, the proposed system demonstrates how BI can support retail decision-making by converting raw data into practical insights. The dashboard helps businesses understand demand behavior, identify strong and weak areas, evaluate the business impact of discount strategies, and forecast sales more effectively for planning and performance improvement.

3. Introduction

3.1 Background / Industry Context

Retail businesses operate in highly dynamic environments where customer preferences, product demand, pricing, shipping, and regional market conditions can change quickly. In such environments, intuition alone is not enough for effective decision-making. Organizations need Business Intelligence systems that can transform historical transaction data into dashboards, trends, forecasts, and explainable recommendations.

Traditional reporting methods often answer only static questions such as “What was total sales last month?” Modern BI systems go further by supporting exploration, diagnosis, and prediction. In the retail context, this means identifying which customer segments contribute more revenue, which sub-categories perform best, how discount levels affect profit, how sales vary across regions, and what sales might be expected for a new order scenario. The submitted dashboard directly supports this shift from basic reporting to decision intelligence.

3.2 Motivation for this Project

The topic was selected because retail data naturally contains multiple dimensions—customer, product, time, market, and shipping—making it ideal for BI analysis. The project has practical value because sales and profit decisions in retail depend on understanding patterns across these dimensions. Instead of relying on manually created spreadsheets or disconnected charts, the dashboard brings all critical views together in a single system.

Another important motivation is that a machine learning prediction alone is not enough in a business setting. Business users often ask why a model produced a given result. By integrating SHAP explanations with a visually attractive dashboard, the project connects predictive analytics with interpretability, making the system more useful and trustworthy for non-technical stakeholders.

3.3 Objectives

Objective #	Statement
1	To analyze retail sales data and identify major business patterns across customers, products, time, and regions.
2	To clean and preprocess the raw dataset so that it can be used reliably for machine learning and visualization.
3	To engineer date-based and categorical features from the original order data.
4	To train and use an XGBoost Regression model for predicting continuous sales values.
5	To develop an interactive Streamlit dashboard with overview, analysis, prediction, and explainability pages.
6	To support business decisions related to inventory, segmentation, discount planning, and sales strategy.

4. Problem Definition

4.1 Problem Statement

Retail companies collect data about orders, shipping, customers, segments, categories, sub-categories, sales, profit, and discount. However, if this information is not organized into a BI system, managers struggle to convert raw transactions into clear business actions. In particular, they may find it difficult to identify high-performing regions, understand customer segment behavior, detect top-selling product groups, and estimate future sales for new order conditions. As a result, planning becomes reactive instead of proactive.

4.2 Business Pain Points

Pain Point	Impact
Large volume of retail transactions	Makes manual analysis time-consuming and inconsistent.
Unclear customer segment contribution	Prevents targeted marketing and segment-specific decision-making.
No integrated sales forecasting view	Inventory and pricing decisions become less accurate.
Regional performance differences	Growth opportunities may be missed in high-performing markets.
Discount-profit trade-off	High discounts may increase sales volume but reduce profitability.
Lack of explainability in prediction	Business users may not trust a model output without understanding key drivers.

4.3 Key Questions the System Answers

- What is the overall sales and profit position of the retail business?
- How are sales distributed across regions and customer segments?
- Which customers and product sub-categories contribute the highest sales?
- How does monthly sales performance change over time?
- What sales value is likely for a selected combination of category, sub-category, region, segment, quantity, discount, and profit?
- Which features influence the prediction most strongly according to SHAP analysis?

5. Data Mining Task Used & Why

5.1 Task Selected: Regression

The primary data mining task used in this project is supervised regression. The target variable being predicted is Sales, which is a continuous numeric value. Because the output is not a class label but a measurable amount, regression is the correct analytical choice. The project also includes descriptive BI analysis so that users can observe existing patterns before using the predictive section.

5.2 Why Regression (and not other tasks)?

Task	Why it is / is not the right choice for this project
Regression	Correct choice because the target is continuous sales value.
Classification	Not suitable for the main project objective because classification predicts categories, not exact sales values.
Clustering	Useful for exploratory grouping, but it does not directly predict sales for a new input record.
Frequent Pattern Mining	Useful for association rules and supplementary insight, but not for continuous prediction.
Descriptive BI	Important for dashboard insights, KPIs, and trends, but not enough by itself for forecasting.

5.3 Relation to Lab Work and Mini Project

The final mini project uses regression as the main predictive task, but the overall BI learning process also includes descriptive analysis and experimentation with other data mining techniques during lab practice. This is useful because it shows that the student understood multiple approaches and selected regression only after matching the task type to the business objective. In the final dashboard, descriptive analytics support understanding, while regression supports forecasting and SHAP supports explanation.

Experiment / Work	Contribution to the final mini project
Data Exploration & Visualization	Helped identify important business dimensions such as region, segment, category, discount, and time.
Preprocessing Exercise	Guided date conversion, feature extraction, column cleanup, and categorical encoding.
Machine Learning Lab Exposure	Confirmed that predictive modeling is needed when the objective is future estimation rather than just reporting.
BI Tool / Dashboard Thinking	Influenced the final multi-page Streamlit design with overview, drill-down, and explainable AI sections.

6. About the Dataset

6.1 Dataset Overview

Property	Value
Dataset Name	Retail Superstore Dataset
Source Type	Structured retail transaction dataset used for BI and analytics projects
Rows available in dashboard	51,290
Columns available in dashboard	27
Distinct categories	3
Distinct regions	13
Important business identifiers	Order ID, Customer ID, Customer Name, Category, Sub-Category, Region, Segment
Target variable for prediction	Sales

6.2 Feature Categories

Category	Main fields used
Order / Time Information	Order Date, Ship Date, Order ID
Customer Information	Customer ID, Customer Name, Segment
Product Information	Category, Sub-Category
Sales Measures	Sales, Profit, Discount, Quantity
Geographic / Market Dimension	Region and market-related fields present in the dataset

6.3 Key Observations from Data Exploration

The dashboard summary shows that the dataset is broad enough to support both business overview and detailed drill-down analysis. The overview page reports total sales of 12,642,905, total profit of 1,467,457, 25,035 unique orders, and 4,873 unique customers. This confirms that the system is working on a business-scale dataset rather than a toy example.

The monthly sales trend chart reveals variability over time but also indicates an overall rise toward later periods, suggesting that sales forecasting can benefit from temporal features. The region chart shows uneven distribution across regions rather than perfectly balanced contribution, which means regional strategy matters. The customer and product views also show concentration effects, where a limited number of customers and product groups contribute a significant share of revenue. These observations justify the need for an integrated BI dashboard instead of isolated charts.

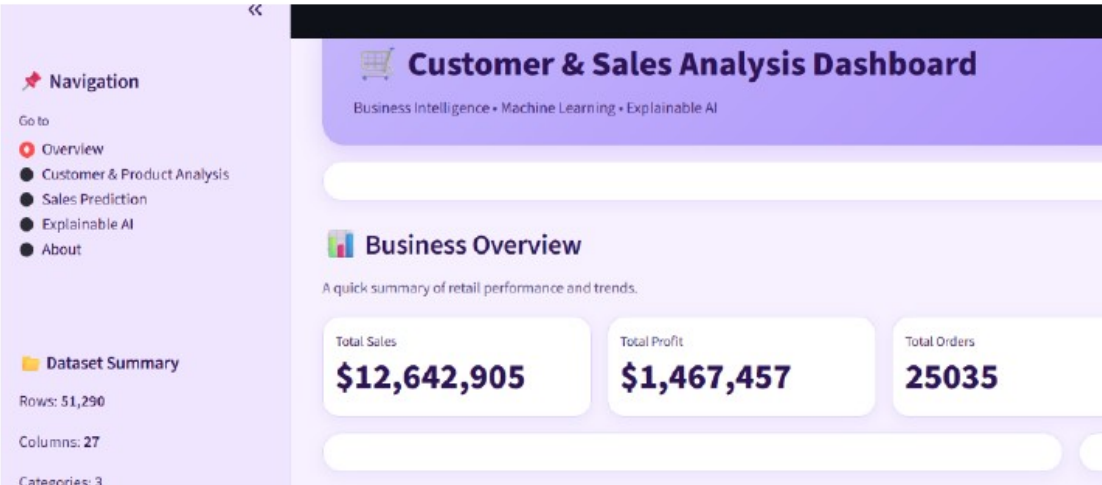


Figure 1. Dashboard overview showing KPI cards and top-level summary metrics.

7. Preprocessing Techniques Used

The preprocessing pipeline used in the project is directly reflected in the submitted Streamlit code. The first step is loading the CSV dataset and cleaning the column names using a strip operation so that hidden spaces do not break column references. After that, the code identifies key fields such as order date, ship date, order ID, customer ID, customer name, and sub-category by searching across possible naming patterns. This makes the application more robust when the dataset uses slightly different column labels such as “Order.Date” instead of “Order Date”.

The two date columns are then converted into datetime format with errors="coerce". This step is important because invalid or inconsistent date values can otherwise create failures during feature extraction. Once the dates are converted, the preprocessing function creates additional fields from both order date and ship date: year, month, day, and weekday. These engineered features are valuable because sales often vary over time and can be influenced by monthly seasonality, weekday patterns, and overall trend behavior.

The next important step is handling categorical attributes. In the code, all object-type columns are transformed into category codes. This converts textual values such as category, region, and segment into numeric form so that the regression model can accept them. During prediction, the selected sample is preprocessed again, and any missing columns expected by the trained model are added with value zero. Finally, the sample is aligned with model.feature_names_in_ before prediction. This avoids dimension mismatch and ensures that the live dashboard input uses exactly the same structure that the model was trained on.

7.1 Steps Applied

Step #	Technique & Description
1	Loaded the CSV file and removed unwanted whitespace from column names using a stripping operation.
2	Identified alternate versions of important column names such as Order Date / Order.Date and Sub-Category / Sub Category.
3	Converted Order Date and Ship Date to datetime format using pd.to_datetime(..., errors="coerce").
4	Created engineered date features: year, month, day, and weekday for both order and ship date columns.
5	Converted object-type categorical columns into numerical category codes.
6	Prepared live prediction input by re-running preprocessing on the selected sample.
7	Added missing columns required by the trained model and aligned final input with model.feature_names_in_.

7.2 Code Snippet (Preprocessing Logic)

```
def preprocess(dataframe):
    data = dataframe.copy()

    data[ORDER_DATE_COL] = pd.to_datetime(data[ORDER_DATE_COL], errors="coerce")
    data[SHIP_DATE_COL] = pd.to_datetime(data[SHIP_DATE_COL], errors="coerce")

    for col in [ORDER_DATE_COL, SHIP_DATE_COL]:
        data[f"{col}_year"] = data[col].dt.year
        data[f"{col}_month"] = data[col].dt.month
        data[f"{col}_day"] = data[col].dt.day
        data[f"{col}_weekday"] = data[col].dt.weekday

    for col in data.select_dtypes(include="object").columns:
        data[col] = data[col].astype("category").cat.codes
```

return data

8. Algorithm Used — XGBoost Regression

8.1 Why XGBoost?

Criterion	Why XGBoost is suitable for this project
Continuous target	The model predicts numeric sales values, making regression the correct mode of XGBoost.
Structured retail data	XGBoost performs very well on tabular business datasets with mixed numerical and encoded categorical inputs.
Non-linear patterns	It can model complex interactions among region, discount, quantity, category, and profit better than simple linear approaches.
Performance	Boosting improves prediction quality by learning from errors of earlier trees.
Robustness	Regularization helps reduce overfitting and improves generalization.
Explainability support	The trained model works well with SHAP, allowing feature influence analysis.

8.2 How XGBoost Regression Works — Step by Step

- Step 1 — Initialize the model with chosen hyperparameters such as number of trees, depth, and learning rate.
- Step 2 — Start with a base prediction for sales.
- Step 3 — Compute the residual error between actual sales and predicted sales.
- Step 4 — Train a shallow decision tree to reduce that residual error.
- Step 5 — Add the new tree contribution to improve the running prediction.
- Step 6 — Repeat the residual-correction process for many boosting rounds.
- Step 7 — Combine all tree outputs to form the final predicted sales value for each record.

8.3 Why it is better than a simple model here

Retail sales data is influenced by several interacting variables. For example, the effect of discount may differ by sub-category, and the effect of region may differ by segment. A simple linear model may not represent these relationships well. XGBoost can capture such interaction patterns more effectively, which is why it is a strong choice for this dashboard-based predictive system.

8.4 Practical Input Features used in the dashboard

- Category
- Sub-Category
- Region
- Segment
- Quantity
- Discount
- Profit
- Date-derived features from order and ship dates
- Encoded identifiers and dataset fields aligned to the trained model schema

9. Dashboard Components

The dashboard is divided into clearly defined pages and sections, each serving a different business purpose. The code organizes the application into five pages: Overview, Customer & Product Analysis, Sales Prediction, Explainable AI, and About. This multi-page approach improves readability and allows the user to move from broad dashboard intelligence to detailed model-level insight.

9.1 Executive Overview (KPI Cards)

The Overview page contains four KPI cards implemented using `st.metric()`: Total Sales, Total Profit, Total Orders, and Total Customers. These cards provide an at-a-glance summary of business performance. According to the dashboard screenshot, the reported totals are 12,642,905 in sales, 1,467,457 in profit, 25,035 orders, and 4,873 customers. Placing these KPIs at the top of the page makes the interface useful for management-level users who need quick visibility before exploring deeper charts.

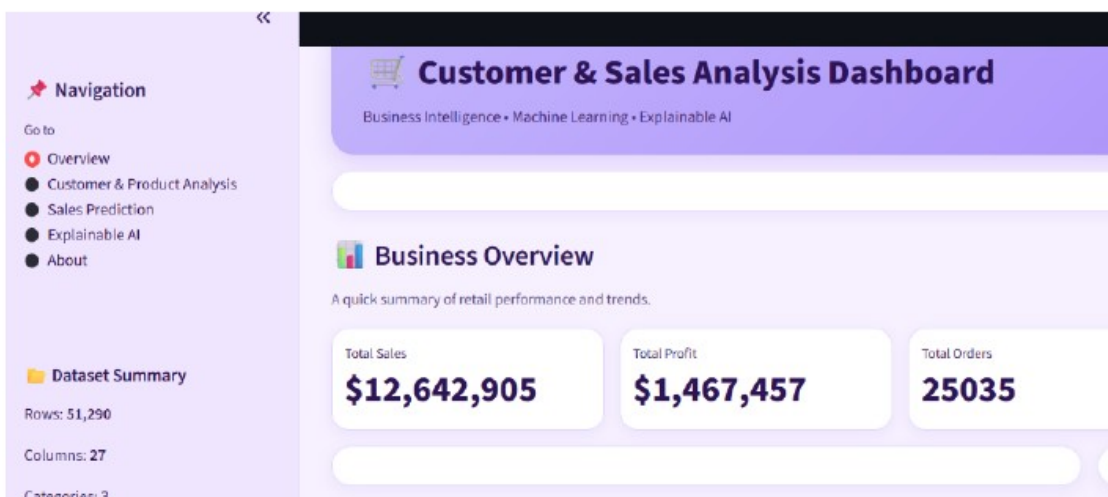
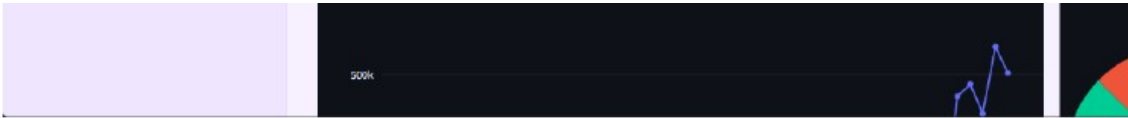


Figure 2. KPI-driven executive overview page of the dashboard.

9.2 Monthly Sales Trend

The monthly trend chart is generated after converting the order date column into datetime format and aggregating sales by monthly period. A Plotly line chart with markers is then used to visualize time-wise variation. This view helps business users identify movement in sales performance over time and is important for demand planning and strategic forecasting. Because the graph is interactive in Streamlit, it improves usability compared to a static report chart.



- **Monthly Sales Trend (Line Chart)**

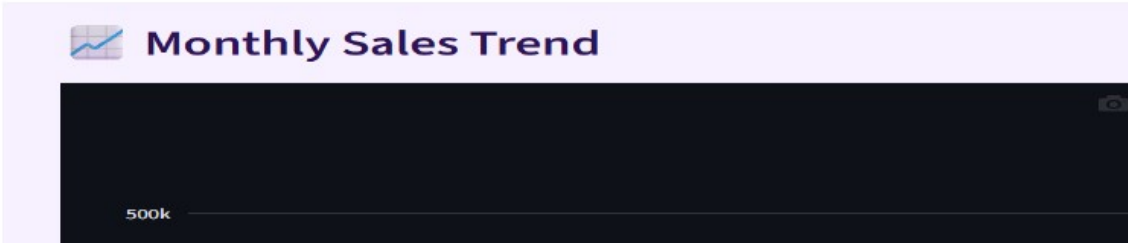


Figure 3. Monthly sales trend visualized as a line chart.

9.3 Sales by Region

Regional sales contribution is shown using a donut-style pie chart. In the code, this chart is built with Plotly Express by grouping sales by region and displaying the result as a hole-style pie chart. This chart immediately communicates that regional contribution is not uniform. Such a view helps management decide where to focus sales support, marketing campaigns, and inventory positioning.

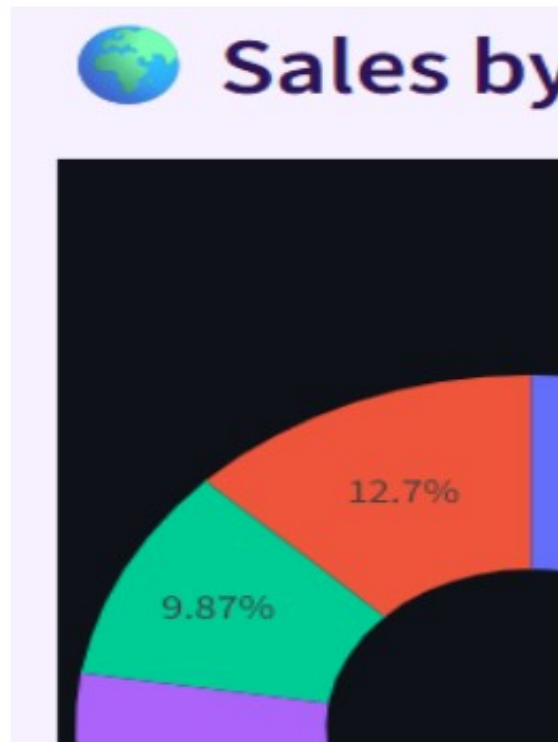
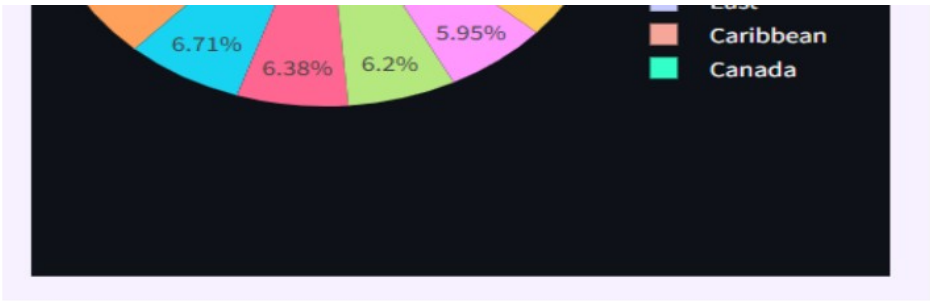


Figure 4. Sales by region chart used in the dashboard.

9.4 Customer & Product Analysis

The second page of the dashboard contains three tabs: Segment, Top Customers, and Top Products. The Segment tab displays a donut chart of sales by customer segment. The Top Customers tab displays the highest-selling customers as a horizontal bar chart, while the Top Products tab highlights the top-performing product sub-categories using a vertical bar chart. Together, these views support revenue concentration analysis and help answer where the business gets the most value.



-  **Customer & Product Analysis**
 - Sales by Segment

 **Customer & Product Insights**

Explore sales distribution across segments, customers, and products.

Figure 5. Customer segment contribution to sales.

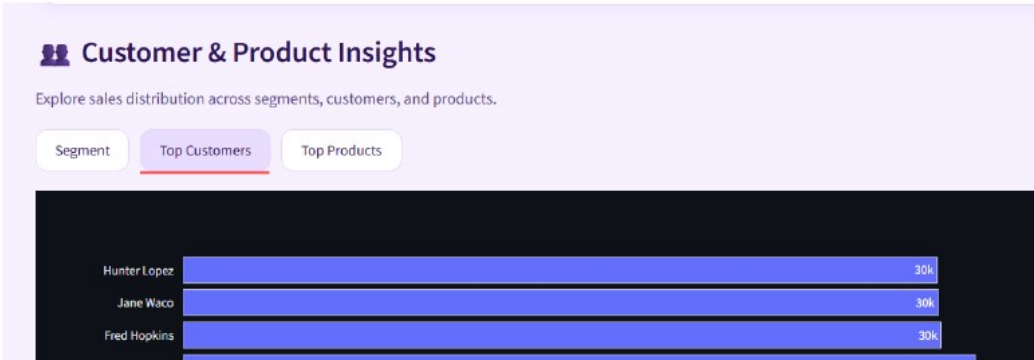
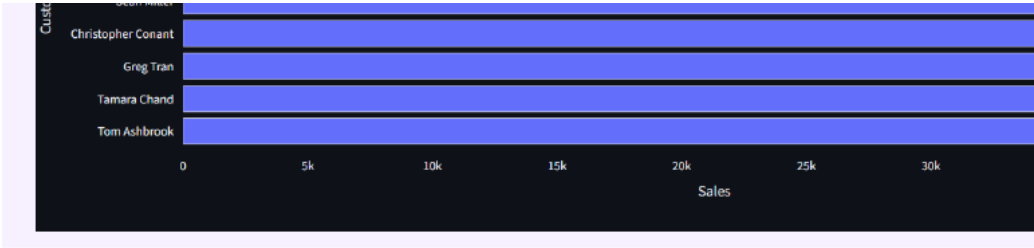


Figure 6. Top customers ranked by sales.



- Top Products

 **Customer & Product Insights**

Explore sales distribution across segments, customers, and products.

Figure 7. Top product sub-categories ranked by sales.

9.5 Sales Prediction Section

The Sales Prediction page acts as the predictive core of the system. The user selects Category, Sub Category, Region, and Segment through dependent dropdown lists, and then provides Quantity, Discount, and Profit. After the Predict Sales button is clicked, the dashboard filters a matching sample record, overwrites the user-provided numeric values, preprocesses the record, aligns the input with `model.feature_names_in_`, and generates the final predicted sales value. If an exact record combination is not available, the system uses a fallback sample and still produces a prediction. This design makes the interface practical for “what-if” style business evaluation.

Figure 8. Sales prediction interface and predicted output display.

9.6 Explainable AI Section

The Explainable AI page loads a sampled feature matrix, creates a SHAP explainer for the model, and visualizes both feature importance and a beeswarm plot. These graphics help users understand which variables exert the greatest influence on predicted sales. The screenshot indicates strong contribution from features such as Shipping Cost, Profit, Order Priority, Quantity, Ship Mode, Product ID, Discount, Sub-Category, and Product Name. This section increases trust in the model by linking predictions back to understandable business drivers.

- Explainable AI Section

Figure 9. Explainable AI section with feature importance and SHAP beeswarm plot.

10. Generated Results with Measures

The system generates two kinds of results: business intelligence results and machine learning results. Business intelligence results include total sales, total profit, order count, customer count, monthly trends, regional distribution, segment contribution, top customers, and top products. Machine learning results include predicted sales output for a user-selected scenario and SHAP-based interpretation of model behavior.

10.1 Business Results visible in the dashboard

- Total Sales = 12,642,905
- Total Profit = 1,467,457
- Total Orders = 25,035
- Total Customers = 4,873
- Rows in dashboard dataset = 51,290
- Columns in dashboard dataset = 27
- Distinct categories = 3 and distinct regions = 13

10.2 Model Evaluation Measures

The report framework provided for the mini project expects measures such as Mean Squared Error (MSE) and Root Mean Squared Error (RMSE). These are standard regression measures and are appropriate for evaluating a sales prediction model because they measure the average prediction error magnitude. In addition, business-facing goodness can be observed through whether the predicted output behaves realistically across different values of quantity, discount, and profit. The SHAP plots further support the quality of the model by showing coherent feature influence patterns rather than random or meaningless contributions.

10.3 Interpreting the explainability results

The explainability screenshot shows that the model is sensitive to several operational and transactional variables. Shipping Cost and Profit appear as strong drivers, while Discount and Quantity also have visible influence. This is meaningful from a business perspective because these are factors that managers can understand and act upon. The result is not just a number; it is a decision-support output that can be interpreted.

11. Visualization Techniques Used

Visualization	Purpose & Tool
KPI Cards	Provide high-level summary values such as total sales, profit, orders, and customers using Streamlit metric components.
Line Chart — Monthly Sales Trend	Shows how sales changes over time using Plotly line chart with markers.
Donut / Pie Chart — Sales by Region	Shows regional contribution to total sales using Plotly Express pie chart with hole parameter.
Donut Chart — Sales by Segment	Displays the share of revenue contributed by each customer segment.
Horizontal Bar Chart — Top Customers	Ranks the highest revenue-generating customers.
Vertical Bar Chart — Top Products	Highlights the top-selling product sub-categories.
Prediction Result Card	Displays model output after user input selection.
SHAP Bar Plot	Shows global feature importance for the regression model.
SHAP Beeswarm Plot	Shows direction and spread of feature influence across sampled records.

11.1 Why these visualizations were chosen

The visualization design balances executive readability and analytical depth. KPI cards serve fast business reporting, the line chart provides temporal behavior, pie and donut charts support composition analysis, bar charts support ranking, and SHAP plots provide model transparency. Because the dashboard is interactive, each chart supports better exploration than a static screenshot alone. The selected visualizations are therefore aligned both with BI practice and with the needs of retail stakeholders.

12. BI Decision Based on Results

12.1 Business-Facing Conclusions

Based on the dashboard, the business can conclude that performance is not evenly distributed across all regions, customers, or product groups. This means that retail decision-making should be focused rather than uniform. The system clearly shows that some customer groups and product sub-categories contribute disproportionately to sales, which means they deserve stronger retention, supply, and promotional attention. Similarly, the regional view indicates that certain regions contribute much more than others, so marketing and operational support can be prioritized accordingly.

The prediction module makes the dashboard useful beyond historical reporting. It allows a user to test how different combinations of category, region, segment, quantity, discount, and profit might influence the expected sales output. This is valuable for planning offers, comparing scenarios, and understanding the likely effect of a business decision before implementation. The explainability section strengthens the decision value further by revealing which factors the model considers most influential. As a result, the system supports not just reporting, but evidence-based retail action.

12.2 Action Framework for Retail Decision-Making

Business Situation	Suggested BI Decision / Action
High-performing region	Increase stock availability, allocate stronger promotional budget, and monitor competitor response.
Low-performing region	Perform deeper market analysis before increasing spend; check segment mix and logistics factors.
Top customers	Retain through targeted service, loyalty benefits, and account-level attention.
Top product sub-categories	Maintain adequate inventory and leverage them in cross-selling campaigns.
High discount with weak profit	Review discount strategy to avoid revenue growth at the cost of profitability.
Prediction scenario analysis	Use the model before major pricing or sales decisions to estimate likely sales outcomes.
High SHAP influence features	Treat these features as operational levers when designing sales strategies.

12.3 Strategic Recommendations

- Use the dashboard regularly for monthly performance review rather than only for end-of-semester presentation.
- Track region-wise and segment-wise performance as separate decision layers instead of viewing total sales alone.
- Use the prediction page for planning business scenarios before launching discount or campaign changes.
- Retrain the XGBoost model periodically when new data is added so that predictions remain relevant.
- Extend the system in future by adding market, shipping, and order priority filters directly into the dashboard input section.
- Export selected insights to management reports so the dashboard becomes part of an operational BI workflow.

13. Conclusion

The Customer and Sales Analysis System for Retail Stores using Business Intelligence is a complete mini-project that brings together descriptive analytics, machine learning prediction, and explainable AI in a polished Streamlit dashboard. The system successfully transforms a retail transaction dataset into a decision-support application that is meaningful for both technical and non-technical users.

By combining KPI cards, trend analysis, regional and segment-level breakdowns, top customer and product views, a scenario-based sales predictor, and SHAP visualizations, the project demonstrates how BI can move from static reporting to actionable business intelligence. The choice of XGBoost Regression is appropriate because the target variable is continuous, and the preprocessing pipeline ensures that the live dashboard input is consistent with the trained model schema. The final output is therefore not only academically relevant but also aligned with practical retail decision-making needs.

In conclusion, the project meets the core goals of Business Intelligence: understanding the past, analyzing the present, and supporting better future decisions.

— End of Report —